[INCH- POUND]
A-A-59405
9 March 1999
SUPERSEDING
MIL-T-21869E
1 December 1989

COMMERCIAL ITEM DESCRIPTION

TRUCKS, LIFT, FORK, ELECTRIC, SIT-DOWN, FRONT WHEEL DRIVE, REAR WHEEL STEER FOR SHIPBOARD USE

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for MIL-T-21869 which is canceled for all federal agencies.

1. SCOPE. This commercial item description covers the general requirements for battery powered, electric, front wheel drive, rear wheel steer, solid-tired type forklift truck for shipboard use.

2. CLASSIFICATION. The forklift trucks shall be of the following classifications as specified (see 7.2).

Type 1 - 4000 pound capacity, EE safety rating

Type 2 - 6000 pound capacity, EE safety rating

Type 3 - 4000 pound capacity, EX safety rating

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Inventory Control Point Code 1041, MHE Division, 5450 Carlisle Pike, P.O. Box 2020, Mechanicsburg, PA 17055-0788.

AMSC N/A FSC 3930

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3. SALIENT CHARACTERISTICS:

3.1 Configuration. This commercial item description covers the types of electric forklifts listed below. In addition to the requirements stated in this CID, each type of forklift truck must be able to meet the following characteristics:

Characteristic	Type 1	Type 2	Type 3
SAFE WORKING LOAD AT 24 IN. LOAD CENTER AND TO FULL LIFT HEIGHT	4000 LB.	6000 LB.	4000 LB.
SAFETY DESIGNATION	EE	EE	EX
FORK LENGTH	36 IN.	36 IN.	40 IN.
FORK SPACING APART(MAX. OUTSIDE DIM.)	38 IN.	40 IN.	40 IN.
FORK SPACING TOGETHER(MAX. OUTSIDE DIM.)	12 IN.	14 IN.	14 IN.
RIGHT ANGLE STACK(MAX.)	146 IN.	160 IN.	152 IN.
OVERALL MAX. LENGTH BETWEEN FORK HEEL FACE AND REAR OF TRUCK	96 IN.	110 IN.	96 IN.
TRUCK WT. (NO LOAD W/BATT.(MAX.)	9900 LBS.	11500 LBS.	11500 LBS.

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Characteristic	Type 1	Type 2	Type 3
BATTERY CAPACITY	850 AMP HRS	850 AMP HRS	765 AMP HRS
CONNECTOR TYPE (MOUNTING HALF ON TRUCK, LOCKING HALF ON BATTERY	EC	EC	N THREE PRONG

3.2 Safety. The forklift shall comply with ASME B56.1, OSHA 29 CFR Chapt XVII Part 1910, UL 583 and as specified herein. The trucks shall bear the UL stamp of approval for both EE and EX trucks. Use of thermoplastic hoses such as SAE 517, 100 R7 and 100R8 is prohibited. In addition, two static straps shall be installeed on each EE truck.

3.3 Motors. Traction motors shall be designed for operation on 36 volt direct current.

3.4 Lubricants. The truck shall operate using only those lubricants that are listed in MIL-HDBK-267.

3.5 Structure. The frame and related structure shall be capable of withstanding 300 per cent of rated load without permanent deformation.

3.6 Forks. The forks and fork carrier shall conform to ASME B56.11.4. Folding forks shall be furnished. Each fork shall be hinged or pivoted near the heel and shall be capable of being folded back manually against the fork upright and the backrest. A manually operated locking device shall hold the fork in the upright position. With no load on the forks, the seated operator shall be able to see at least one fork tip at all lift heights and fork positions.

3.7 Load backrest extension. A 48 inch removalble load backrest extension shall be provided.

3.8 Tires. The tires shall be solid type and shall be selected from sizes listed in the Tire and Rim Association (T&RA) Yearbook. Tires shall be lug type with non-directional treads and smooth tires are not acceptable. Steer tires shall be siped and there shall be two sipes on each lug. Tire loadings shall not exceed the values listed in the T&RA Yearbook.

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Drive wheel tires shall be 18X8X12.125 minimum and steer tires shall be 16X6X10.5 minimum to avoid deck overloading.

3.9 Steering. Power steering shall be provided. If a tilt steering wheel is provided, there shall be adequate clearance between the steer wheel in its full forward tilt position and the mast in its full rearward tilt position for an operator's gloved hand.

3.10 Service brakes. The brake system shall conform to ASME B56.1. Four wheel brakes shall be provided. The master cylinder shall be dual output type providing a dual hydraulic circuit(split hydraulic) brake system. Non-asbestos brake linings shall be used.

3.11. Travel disconnect. A travel disconnect system shall be provided such that upon removal of the operator's weight (85 lbs (39 kg) min.) from the seat, the drive wheel brakes will be applied and the current to the traction motors will be interrupted. This control shall not be deactivated with the return of the operator's weight on the seat. This control is to reset after the master switch is turned "on" and the direction control is moved to either the neutral position or the forward/reverse direction. This device shall hold the loaded vehicle on a 26.8 per cent slope.

3.12 Parking brake. A hand or foot operated parking brake with a locking device shall be provided in addition to the travel disconnect system.

3.13 Seat. A cushion seat and cushion backrest, adjustable forward and reverse shall be provided.

3.14 Towing device. A pin type towing device shall be provided in the center rear of the truck. The towing device shall accept a lunette conforming to MS 51336 without binding. The towing device shall not protrude beyond the path followed by the rear end of the truck when making a minimum radius turn.
The towing device shall be capable of withstanding the force induced during drawbar drag test in ASME B56.1.

3.15 Slinging and tie down provisions. Permanently affixed slinging and tie down provisions conforming to MIL-STD-209, Classes 1 and 2 or Class 3 for type II and III excluding air transport requirements, shall be provided.

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3.16 Side shift. Side shift attachments shall be provided to hydraulically provide a minimum movement of 4 inches on each side of center with rated load.

3.17 Weighing device. Each truck shall be equipped with a weighing device to measure the amount of weight on the forks. The device shall be capable of weighing a minimum of 100 pounds(46 kg) and a maximum weight equal to rated load plus 300 pounds(138 kg). Weighing device shall be capable of being adjusted for 25 percent tare. The weighing device shall be capable of withstanding pressure caused by placing 150 percent of truck rated load on the forks without sustaining damage. Readout shall be readable by the operator when in the seated position.

3.18 Master switch. A key operated master switch shall be provided such that its removal deactivates all electrical control circuits. All trucks under each contract shall be keyed identically. Two keys shall be supplied with each truck.

3.19 Hour meter. An electrically operated hour meter which registers total hours of operation of the traction and pump motors shall be provided.

3.20 Flood light. Each truck, except EX type, shall be provided with two shock mounted, sealed beam, 25 watt minimum flood lamps. One shall be mounted for forward illumination and one for rearward illumination. Floodlight shall have vertical and horizontal adjustment and shall be provided with a protective guard to prevent impact damage. Each floodlight shall have a set of three lenses. Color of lenses shall be clear, red and yellow. red and yellow lenses shall be stored in a pocket located on the back of the operator's seat.

3.21 Stop/tail light. One automotive, red, reflector-type combination, stoplight and tail lamp of not less than 15 candle power shall be provided and shall be mounted on the rear end of the truck within the plan outline of the truck. Type EX trucks shall not be provided with stop/tail lights.

3.22 Horn. Truck shall be equipped with an electric horn except type EX truck shall have an air horn.

3.23 Battery protector. Each truck shall be equipped with a battery protector to protect the battery from approaching a low state of charge

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3.24 Lifting speed. Minimum lifting speed over the entire distance from ground to maximum fork height shall be 25 feet per minute.

3.25 Lowering speed. Minimum lowering speed of unloaded forks from maximum fork height to ground level shall be shall be 27 feet per minute.

3.26 Speed. While carrying rated capacity on level surface, trucks shall be capable of attaining 3.5 to 4.5 miles per hour in forward and reverse directions.

3.27 Slope ascension. Trucks shall be capable of ascending a grade of 26.8 percent in the forward direction with rated load.

3.28 Upright tilt. When loaded, truck masts shall have forward tilt of 2 degrees plus or minus 0.5 degrees and rearward tilt of 6.5 degrees plus or minus 0.5 degrees.

3.29 Collapsed mast height. Maximum collapsed mast height of unloaded trucks shall be 68 inches.

3.30 Lift height. The minimum fork height of truck bearing rated capacity load and the uprights fully extended vertically, measured from the ground to the load carrying surface of the fork at the rated load center shall be not less than 100 inches.

3.31 Free lift height. Lift of forks without increase in specified collapsed mast height shall be not less than 42 inches.

3.32 Under clearance. The truck with rated load shall have sufficient Under clearance to permit safe operation from one horizontal plane to another using a 26.8 per cent slope. Under clearance beneath the mast assembly, when in a true vertical position and with rated load on the forks shall not be less than 2.5 inches.

3.33 Lift assembly drift. The lift assembly shall be capable of holding the rated load at maximum lift height for 10 minutes with not more than 1.75 inches of vertical drift and not more than 1 degree rotational drift from the vertical when hydraulic fluid temperature at the inlet to each cylinder is not less than 120 deg. F.

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3.34 Stability. The truck carrying rated load shall have longitudinal stacking stability on a 6 percent slope, longitudinal traveling stability on a 26.8 percent slope, lateral stacking stability on a 9 percent slope and lateral traveling stability on a 40 percent slope.

3.35 Overhead guard. The maximum overhead guard height shall be 91 inches. No truck component such as lights shall be mounted on the overhead guard.

3.36 Stopping distance. Truck with rated load shall be capable of stopping from a rearward maximum speed in the distance indicated in ASME B56.1.

3.37 Deck loading. The truck's tire loading with rated load and forks at each maximum sideshift position shall not exceed the weight listed on Table 1 in both stationary and transporting modes; and, in no case, shall the deck loading exceed 250 psi.

3.38 Temperature. The truck shall be capable of continuous operation from 0 deg.F.(-18 deg C.) to 120 deg. F.(49 deg. C.).

3.39 Resistance to saline atmosphere. All truck components shall withstand the corrosive effects of salt fog in accordance with MIL-STD-810 Method 509.2 Salt Fog without any loss of performance of the truck.

3.40 Power consumption. The truck shall be able to be cycled continuously for 7.5 hours with one battery only and no recharging is allowed.

3.41 High Impact shock. The truck and the battery as a component of the truck shall withstand a type A high shock as a grade A item in accordance with MIL-S-901.

3.42 Electromagnetic interference control. The trucks shall be equipped for control of electromagnetic interference in accordance with MIL-STD-461. For radiated emissions, procedure RE 102 shall be used with limits shown in figure RE 102-1 over a frequency range of 10 KHZ to 1 GHZ. For susceptibility to electric fields, procedure RS 103 shall apply with a frequency range from 10 KHZ to 1 GHZ at a limit level of 10 Volts/Meter.

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3.43 Rain. The truck shall be capable of performing all its normal operations after being subjected to a rainfall of not less than four (4) inches per hour for a period of one hour.

3.44 Reliability. Truck shall be able to complete 2000 laps around an endurance course without failure.

3.45 Treatment and painting. All exterior surfaces of the truck shall be thoroughly cleaned, primed and painted in accordance with standard commercial practice. The finish coat shall be yellow color 13538 of FED-STD-595.
Paint shall be lead free and chromate free. The mast and overhead guard may be painted black.

3.46 Information plates. The truck shall be equipped with the following information plates conforming to A-A-50271, composition C. Plates shall be furnished and installed by the contractor.

3.46.1 Identification plate. Each forklift shall be equipped with an identification plate that shall include truck nomenclature such as, "TRUCK, LIFT, FORK, ELECTRIC, SOLID TIRES", truck model number, truck serial number, contract number, lifting capacity, lift height, USN registration number, vehicle weight, shipping weight, cube dimension, name or stamp of government inspector, date shipped, technical manual stock number, high shock test date, warranty expiration date and manufacturer's name and address.

3.46.2 Shipping data plate. Each forklift shall be equipped with a shipping data plate that shall show silhouette of the forklift in transport position indicating the center of gravity and the location and capacity of the lifting and tie down attachments.

3.46.3 Instruction plates. Each forklift shall be equipped with instruction plates including warnings, cautions and diagrams describing any special or important procedures to be followed in assembling, operating or servicing the forklift. Commercially proven decals may be substituted for these instruction plates.

3.46.4 Wheel loading plate. Each forklift shall be equipped with a wheel loading plate to show wheel loading on each wheel when the forklift is carrying the rated load and when the forklift is unloaded.

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3.46.5 Safety designation plate. Each forklift shall be equipped with a safety designation plate to show the truck manufacturer's name, the truck's complete model number, the safety designation (Type EE or EX), independent testing laboratory's name, independent testing laboratory's registration or index number assigned to the inspected truck, and the date and location of the independent laboratory's inspection.

3.46.6 Battery plate. Each forklift shall be equipped with a battery plate to show the battery weight, dimensions, ampere-hour capacity at a 6-hour rate, voltage, number of cells, number of plates, cable lead length and position.

3.47 Body marking. The USN number shall be indelibly stamped into the external plate of the truck body on the right side of the truck approximately 6 inches below the center of the entrance to the operator's compartment.
Marking shall be in 0.25 in high block letters and numbers.

3.48. Vehicle marking. The vehicle shall be marked in black block letters, unless otherwise indicted, as follows.

3.48.1 USN registration number. The assigned USN registration number shall be painted in block in numbers 3 inches high on each side of the truck.

3.48.2 Vehicle weight. The words, "VEHICLE WEIGHT" followed by the forklifts's empty weight shall be painted on each side and rear of the truck mast in numbers 2 inches high. White numbers may be used when the mast is painted black.

3.48.3 Safe working load. Each truck shall be marked with the letters "SWL" followed by the truck's rated lifting capacity on each side of the mast in 2 inch high letters and numbers. White numbers and letters may be used when the mast is painted black.

3.48.4 Safety designation. The safety designation (EE or EX) shall be painted on each side and the rear of the truck in accordance with UL 583.

3.48.5 Slinging and tie down. Slinging and tie down markings shall be in accordance with MIL-STD-209.

3.48.6 No rider. "No Rider" warning shall be painted in 2 inch letters on the rear of the mast.

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3.48.7 Paint. The words "Painted with lead free and chromate free paint on (MM/YY) by (Contractor Name)" shall be painted in minimum 1/4 inch high letters on each side of the truck.

3.48.8 Brake linings. The words "Non-asbestos brake linings used. Replacements to be non-asbestos." shall be painted in 1 inch high letters on the truck.

3.48.9 Shipboard use. Each truck shall be marked with the words "SHIPBOARD USE APPROVED" in 1 1/2 inch high letters on each side and rear of the truck. Batteries shall be marked with the words "SHIPBOARD CERTIFIED" in a conspicuous place on the battery case in 1 inch high letters.

3.48.10 Structural testing. Each truck shall be marked with the words "STRUCTURALLY TESTED", the test date and the manufacturer's name in 1 1/2 inch high letters on each side of the truck.

3.48.11 EX truck. For all EX safety rated trucks, a single 6 inch wide blue diagonal stripe shall be painted at a 45 degree angle across the right side , left side and rear of the truck.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the forklift.

5. QUALITY ASSURANCE PROVISIONS.

5.1 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

6. PACKAGING. Preservation, packaging , packing and marking for shipment shall be specified in the contract or order.

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7. NOTES.
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

7.1 Referenced documents.

Government Documents.

Specifications and standards and handbooks. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation, (see 7.2).

SPECIFICATIONS

FEDERAL

A-A-50271 - Plate, Identification

DEPARTMENT OF DEFENSE

MIL-S-901 - Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment, and Systems, Requirements for

STANDARDS

FEDERAL

FED-STD-595 - Colors

DEPARTMENT OF DEFENSE

MIL-STD-209 - Slinging Eyes and Attachments for Lifting and Tying Down Military Equipment

MIL-STD-461 - Electromagnetic Emission Susceptibility Requirements for the Control of Electromagnetic Interference

MIL-STD-810 - Environmental Test Methods

MS 51336 - Lunette-Coupler, Drawbar, Ring

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DEPARTMENT OF DEFENSE

MIL-HDBK-267(SH)-Guide for Selection of Lubricants and Hydraulic Fluids for use in Shipboard Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the DODSSP, Standardization Document Order Desk, 700 Robbins Avenue, Philadelphia, PA 19110-5094.)

Other Government documents, drawings and publications. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF LABOR(DOL)

29 CFR, Part 1910 - Occupational Safety and Health Standards (OSHA)

(Application for copies should be made to a local DOL officer or the Superintendent of Documents , U. S. Government Printing Office, Washington DC 20402)

Non-Government Publications. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 7.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS(ASME)

B 56.1 - Safety Standard For Low Lift and High Lift Trucks
B56.11.4 - Forks and Fork Carriers for Powered Industrial Trucks

(Applications for copies should be addressed to: American Society of Mechanical Engineers, 345 East 47th Street, New York,NY 10017)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J517 - Hydraulic Hose

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(Application for copies should be addressed to: Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, Pennsylvania 15096).

THE TIRE AND RIM ASSOCIATION, INC.(T&RA)

Tire and Rim Association Yearbook

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 W. Market St., Akron, OH 44313)

UNDERWRITERS LABORATORIES, INC.(UL)

UL 583- Standard for Electric-Battery-Powered Industrial Trucks

(Application for copies should be addressed to: Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

7.2 Ordering data. Acquisition documents shall specify the following:

(a) Title, number and date of this Commercial Item Description.

(b) Type required. (see 3.1)

(c) Issue of the DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced.

(d) When a government first article inspection is required, location where user test is to be conducted and required government clearances.

(e) When a technical manual verification review is required, location where review is to be conducted and required government clearances.

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7.3 Technical Manuals. The requirements for technical manuals should be considered when this commercial item description is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

7.3.1 Technical manual verification review. When a government technical manual verification is required (see 7.2), the government will conduct the review as specified herein.

7.3.1.1 Technical manual verification review. The technical manual verification review will be conducted by the government to validate that the truck's final technical manual accurately reflects the truck configuration, general information, operating instructions, safety rating, preventative maintenance, repair and overhaul, illustrated parts breakdown and parts list index. The truck will be operated only to the extent required by the technical manual to ensure that each section of the manual is accurate. Repair and overhaul operations will not be conducted, but will be reviewed for clarity. The illustrated parts breakdown and parts list index will be reviewed to the extent possible without major disassembly of the truck. The manual review shall be conducted after successful completion of first article inspection. The truck shall be available for review for a period of fourteen consecutive calendar days. Government review will be conducted in accordance with instructions in the contract or purchase order. In the event the technical manual does not accurately reflect the truck, a written list of discrepancies will be included in the prefinal technical manual review comments.

7.3.1.2 Delivery. The contractor shall deliver one preproduction truck to the Naval facility specified herein (see 7.2). The cost of transportation to and from the test site shall be borne by the contractor. The contractor shall provide notification of truck shipment to the contracting officer and the Naval facility a minimum of 20 days prior to shipment.

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7.3.1.3 Contractor personnel. When requested by the contractor and approved by the government, contractor's personnel may witness the review. When required (see 7.2), government clearance for contractor personnel will be provided.
The cost of transportation and billeting contractor personnel shall be borne by the contractor.

Custodian:

Navy -SA

Review Activity:

Navy - SH

Preparing Activity:

Navy - SA

(Project No. 3930-0694)

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